



Francisco Samagaio

Junior Data Engineer
Porto, Portugal

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Python | SQL | ETL Pipelines | Data Warehousing

Work Experience

Junior Data Analyst — Natixis in Portugal

Dec 2024 – Present

- Developed and optimized SQL queries in Apache Hive supporting approximately 300 financial reports within a data warehouse environment.
- Built and maintained automated data pipelines and reporting workflows in EasyMorph for Corporate & Investment Banking teams.
- Migrated approximately 80 reports from IBM Cognos to EasyMorph, adapting and optimizing SQL logic for the target platform.
- Designed, maintained, and validated 30+ tables supporting ETL pipelines and financial reporting workflows.

Intern Developer — Natixis in Portugal

Dec 2023 – Dec 2024

- Developed Python automation to replace legacy PowerShell workflows and reduce manual data processing.
- Built ETL pipelines using Python, Pandas, and NumPy to integrate and transform 150+ datasets across CSV, Excel, TXT, and JSON formats.
- Used Git and Linux to support development and maintenance of data processing workflows.
- Supported financial batch processing systems using COBOL, JCL, and CICS.

Research Intern / Master's Thesis — Fraunhofer Portugal AICOS

Feb 2023 – Jul 2023

- Developed a generative AI research project for ceramic color generation, using Python and Linux-based HPC environments for model training and experimentation.
- Built data preparation and evaluation workflows, assessing model outputs using quantitative metrics.

Technical Skills

Programming: Python, SQL, C/C++, Bash

Data Engineering: ETL Pipelines, Data Integration, Data Warehousing, Batch Processing

Data Processing: Pandas, NumPy, Data Cleaning, Data Transformation

Tools & Platforms: Git, Linux, Jupyter, EasyMorph

Databases & Query Engines: Apache Hive, SQL Query Optimization

Foundational Knowledge: PySpark, Medallion Architecture

Currently Learning: Databricks

Education

University of Porto — Faculty of Engineering | 2018–2023

- Master's Degree in Electrical and Computer Engineering
- Thesis: Generative AI for Ceramic Color Generation

University of Genoa | 2022

- Erasmus+ Exchange Program

Languages

Portuguese — Native · **English** — Fluent · **Spanish** — Intermediate · **French** — Beginner